

Object Tracking Application

A powerful and user-friendly web application for tracking moving objects in videos using computer vision techniques. Built with Python, OpenCV, and Streamlit.

Python 3.7+ OpenCV 4.0+ Streamlit 1.0+

Overview

This application uses advanced background subtraction algorithms to detect and track moving objects in video files. It provides real-time visualization, customizable detection parameters, and comprehensive statistics about detected objects.

Features

Core Functionality

- **Real-time Object Detection:** Tracks moving objects using MOG2 background subtraction
- **Bounding Box Visualization:** Draws boxes around detected objects with area labels
- **Multiple Video Format Support:** MP4, AVI, MOV, WMV, MKV

User Interface

- **Professional GUI:** Clean, intuitive interface with custom styling
- **Wide Layout:** Optimized for viewing videos and statistics
- **Responsive Design:** Adapts to different screen sizes

Customization Options

- **Detection Parameters:**
 - Adjustable minimum object area (100-2000 pixels)
 - Background learning rate control (0.001-0.1)
- **Display Options:**
 - Custom bounding box colors
 - Adjustable box thickness (1-5 pixels)
 - Toggle detection mask view
 - Real-time statistics display
- **Playback Controls:**
 - Variable playback speed (0.25x to 2.0x)

Analytics & Statistics

- **Video Information:** Resolution, FPS, frame count, duration
- **Real-time Metrics:**
 - Current frame number
 - Objects detected in current frame
 - Total detections across video

- Processing FPS
- **Processing Summary:**
 - Total frames processed
 - Total detections
 - Maximum objects in a single frame
 - Average detections per frame

Advanced Features

- **Noise Reduction:** Morphological operations to filter false detections
- **Shadow Detection:** Built-in shadow detection in background subtraction
- **Progress Tracking:** Visual progress bar with percentage completion
- **Side-by-side View:** Optional mask visualization alongside original video

Installation

Prerequisites

- Python 3.7 or higher
- pip package manager

Setup Instructions

1. Clone or download this repository

```
cd object_tracking_app
```

2. Install required dependencies

```
pip install -r requirements.txt
```

If `requirements.txt` doesn't exist, install manually:

```
pip install streamlit opencv-python numpy
```

Usage

Running the Application

1. Start the Streamlit server

```
streamlit run ObjectTracking.py
```

2. Open your web browser

- The application will automatically open at <http://localhost:8501>
- If it doesn't open automatically, navigate to the URL shown in the terminal

Using the Application

1. Upload a Video

- Click the "Upload your video file" button
- Select a video file (MP4, AVI, MOV, WMV, or MKV)

2. Adjust Settings (Optional)

- Open the sidebar to customize detection parameters
- Adjust minimum object area to filter small movements
- Modify learning rate based on your video characteristics
- Change bounding box color and thickness
- Toggle detection mask view

3. Process Video

- The video will automatically start processing after upload
- Watch real-time detection with bounding boxes
- Monitor statistics in the dashboard

4. Review Results

- After processing completes, view the comprehensive summary
- Analyze total detections and average objects per frame

Parameter Guide

Minimum Object Area

- **Range:** 100-2000 pixels
- **Default:** 300 pixels
- **Purpose:** Filters out small noise and focuses on larger objects
- **Recommendation:**
 - Increase for large objects (vehicles, people)
 - Decrease for small objects (small animals, objects)

Background Learning Rate

- **Range:** 0.001-0.1
- **Default:** 0.01
- **Purpose:** Controls how quickly the background model adapts
- **Recommendation:**
 - Higher values (0.05-0.1): Dynamic backgrounds, changing lighting
 - Lower values (0.001-0.02): Static backgrounds, stable lighting

Playback Speed

- **Options:** 0.25x, 0.5x, 0.75x, 1.0x, 1.5x, 2.0x
- **Default:** 1.0x
- **Purpose:** Control processing speed for better visualization

Project Structure

```
object_tracking_app/  
├── ObjectTracking.py    # Main application file  
├── README.md           # Project documentation  
└── requirements.txt     # Python dependencies (optional)
```

Technical Details

Algorithm

- **Background Subtraction:** MOG2 (Mixture of Gaussians)
- **Contour Detection:** External contours with simple approximation
- **Noise Reduction:** Morphological opening and closing operations
- **Shadow Detection:** Enabled in MOG2 algorithm

Libraries Used

- **Streamlit:** Web application framework
- **OpenCV (cv2):** Computer vision operations
- **NumPy:** Numerical operations
- **tempfile:** Temporary file handling
- **datetime:** Timestamp operations

Tips for Best Results

1. Video Quality

- Use videos with relatively static backgrounds
- Ensure good lighting conditions
- Avoid excessive camera shake

2. Parameter Tuning

- Start with default values
- Adjust minimum area based on object size
- Increase learning rate for outdoor videos with changing light
- Decrease learning rate for controlled indoor environments

3. Performance

- Larger videos may take longer to process
- Adjust playback speed if processing is slow
- Close other applications for better performance

Troubleshooting

Video Won't Load

- **Solution:** Ensure the video format is supported (MP4, AVI, MOV, WMV, MKV)
- **Solution:** Try re-encoding the video with standard codecs

Too Many False Detections

- **Solution:** Increase the minimum object area
- **Solution:** Decrease the background learning rate

Missing Detections

- **Solution:** Decrease the minimum object area
- **Solution:** Increase the background learning rate

Slow Processing

- **Solution:** Use a smaller resolution video
- **Solution:** Close unnecessary applications
- **Solution:** Increase playback speed parameter

License

This project is open source and available for educational and personal use.

Contributing

Contributions, issues, and feature requests are welcome! Feel free to improve the application.

Contact

For questions or feedback, please open an issue in the repository.

Acknowledgments

- OpenCV community for computer vision algorithms
- Streamlit team for the amazing web framework
- Background subtraction research community

Made with  using Python, OpenCV, and Streamlit